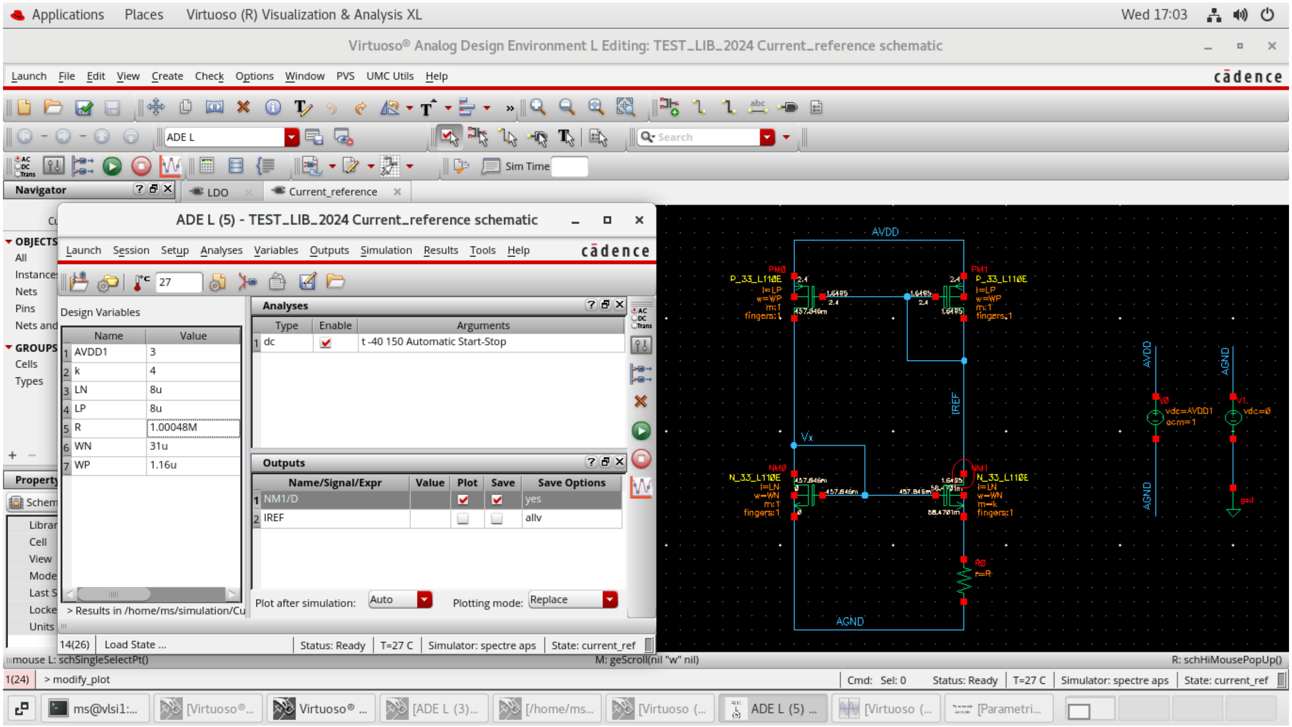
MALOBI MUKHERJEE
ANALOG POWER IC DESIGN ASSIGNMENT-1
MS2023006

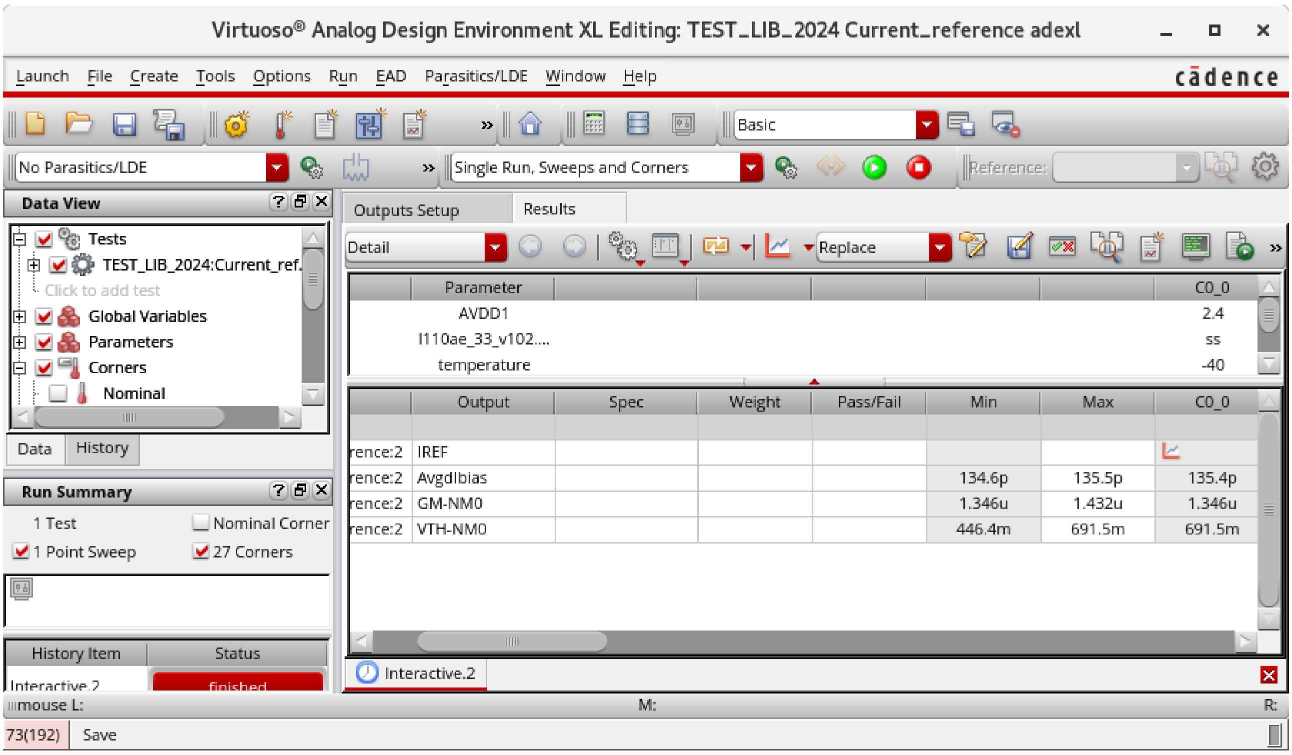
Question – 1:



Schematic

Given:- VDD = 3V, k = 4, Ibias = 100n A
Calculated:- Rbias = 1.00048M Ohm, WN = 0.31um, WP = 1.16um, LN = 8um, LP = 8 um

Corners – C0 (SS), C1 (TT), C2 (FF)



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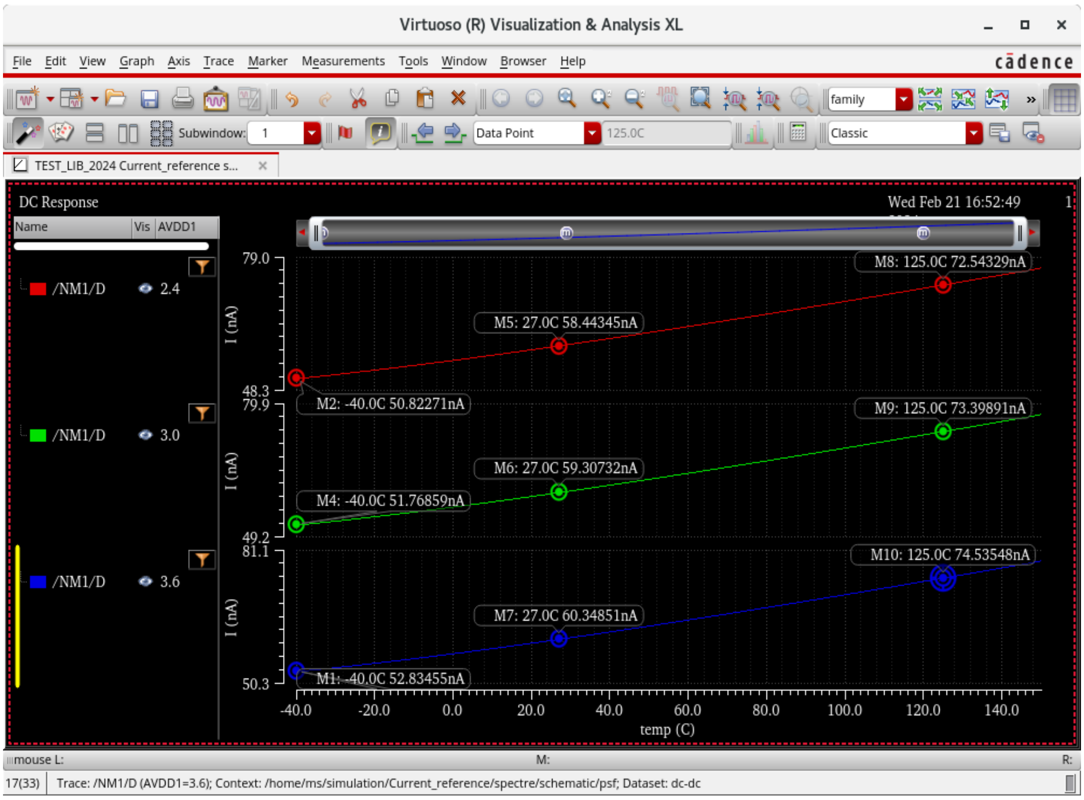

<u>Iref vs Temperature</u>

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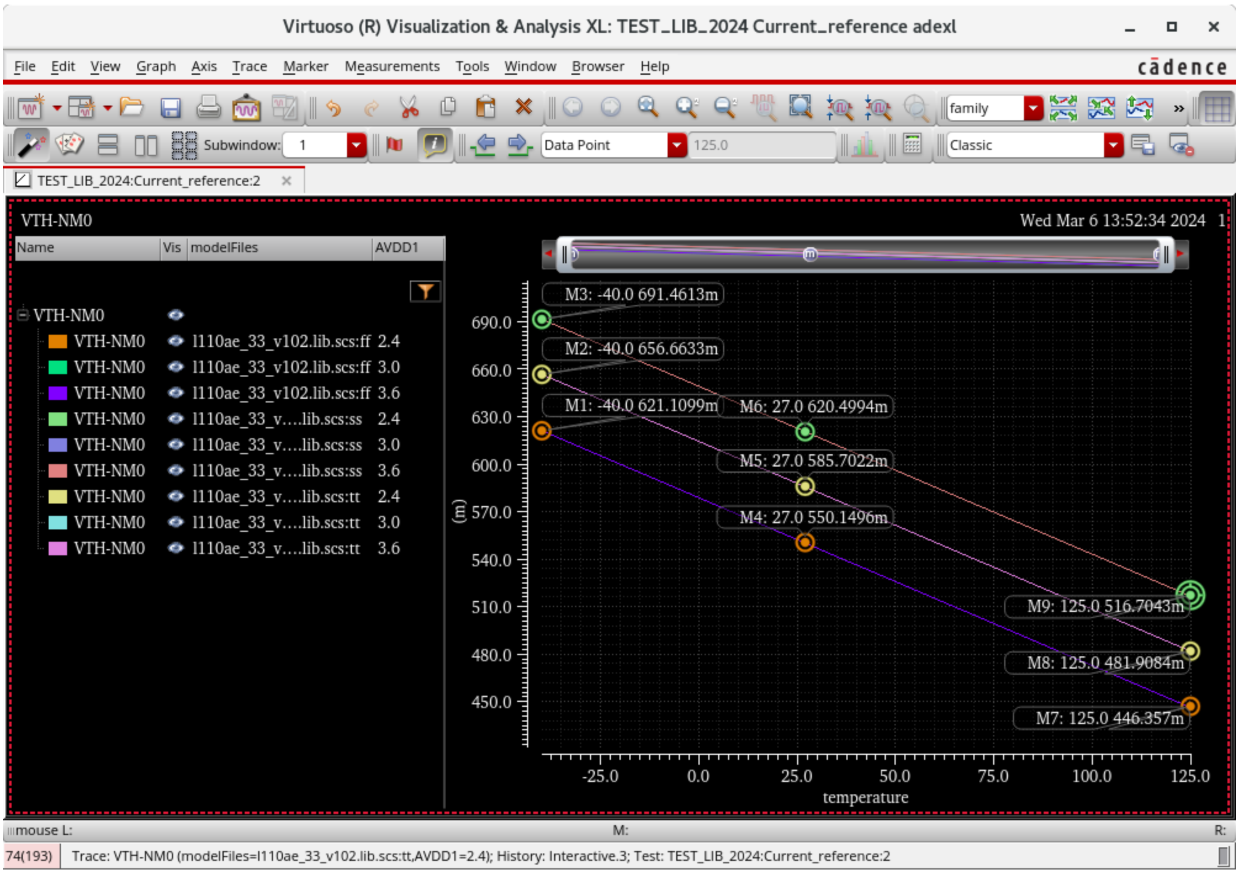
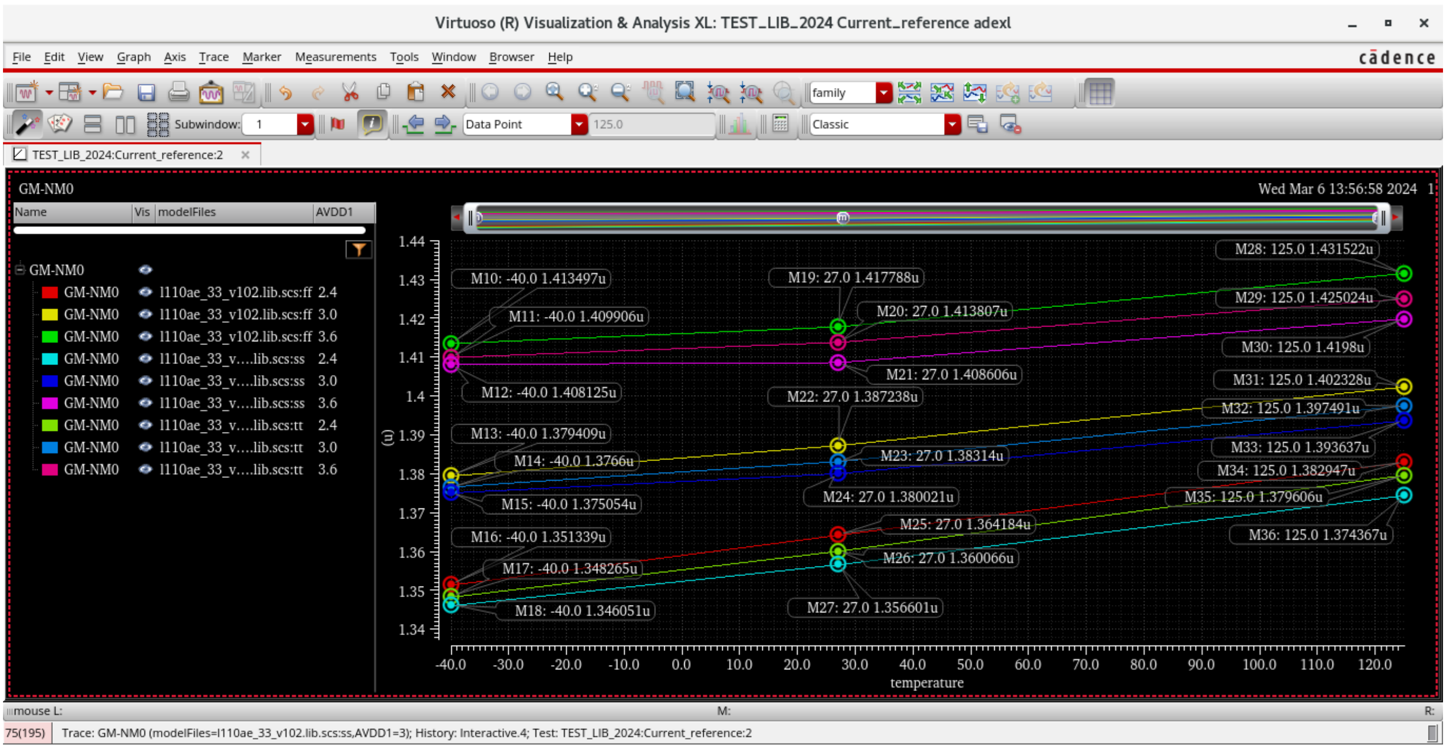


Vth-NM0 vs Temperature



Gm-NM0 vs Temperature

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Corners – C0 (SS), C1 (TT), C2 (FF)

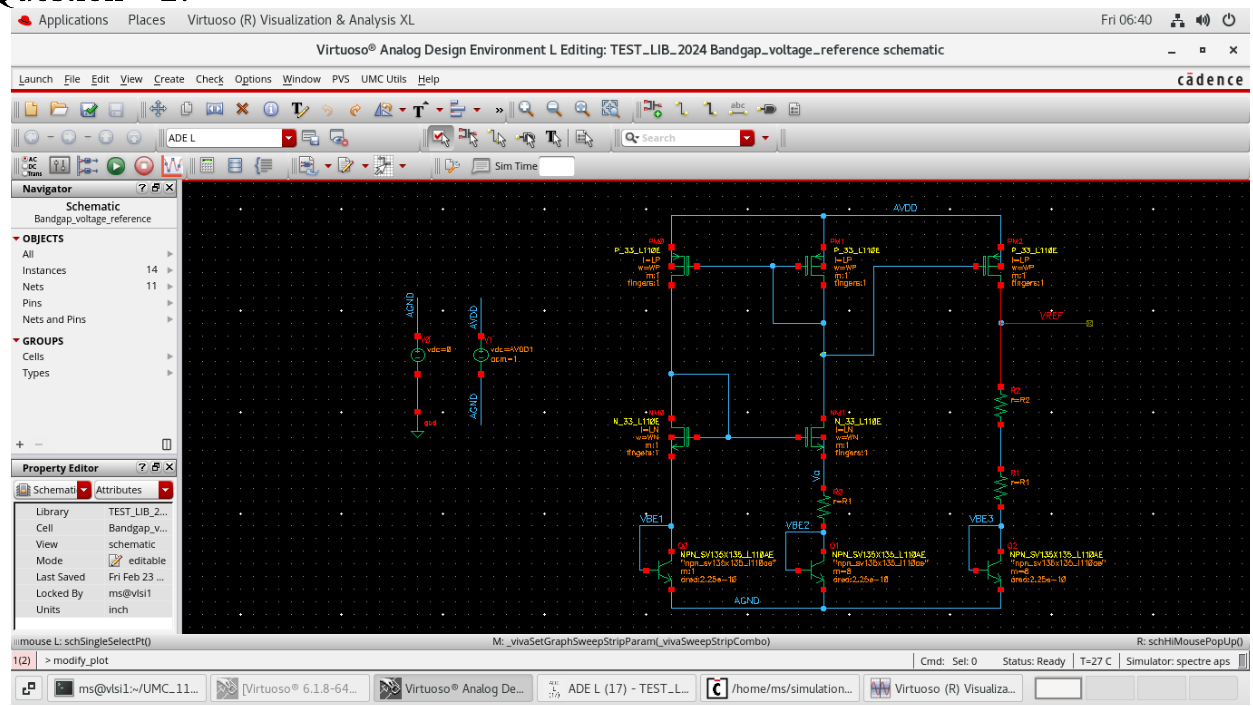
Corners		SS			FF			TT		
AVDD		2.4V	3.0V	3.6V	2.4V	3.0V	3.6V	2.4V	3.0V	3.6V
Vth-NM0 (V)	@-40C	691.4m	691.4m	691.4m	621.11m	621.11m	621.11m	656.66m	656.66m	656.66m
	@27C	620.499m	620.499m	620.499m	550.15m	550.15m	550.15m	585.702m	585.702m	585.702m
	@125C	516.704m	516.704m	516.704m	446.357m	446.35m	446.357m	481.908m	481.908m	481.908m
Gm-NM0 (S)	@-40C	1.346u	1.375u	1.408u	1.35u	1.37u	1.4135u	1.348u	1.376u	1.409u
	@27C	1.356u	1.38u	1.408u	1.364u	1.38u	1.417u	1.36u	1.38u	1.413u
	@125C	1.374u	1.393u	1.419u	1.38u	1.402u	1.431u	1.37u	1.39u	1.425u
Ibias (A)	@27C	58.962n	59.84n	60.86n	57.9n	58.84n	59.91n	58.44n	59.30n	60.34n
avg(dIbias/dT)	pA/C	133.05	127.93	132.01	127.25	126.3	127.19	127.81	130.86	128.94

Vth-NM0 decreases with increase in temperature.

Gm-NM0 = sqrt(2*Iref*beta)

From the above graph,it is seen that Iref increases with the increase in temperature, and so does the Gm-NM0.

Question – 2:-



Schematic

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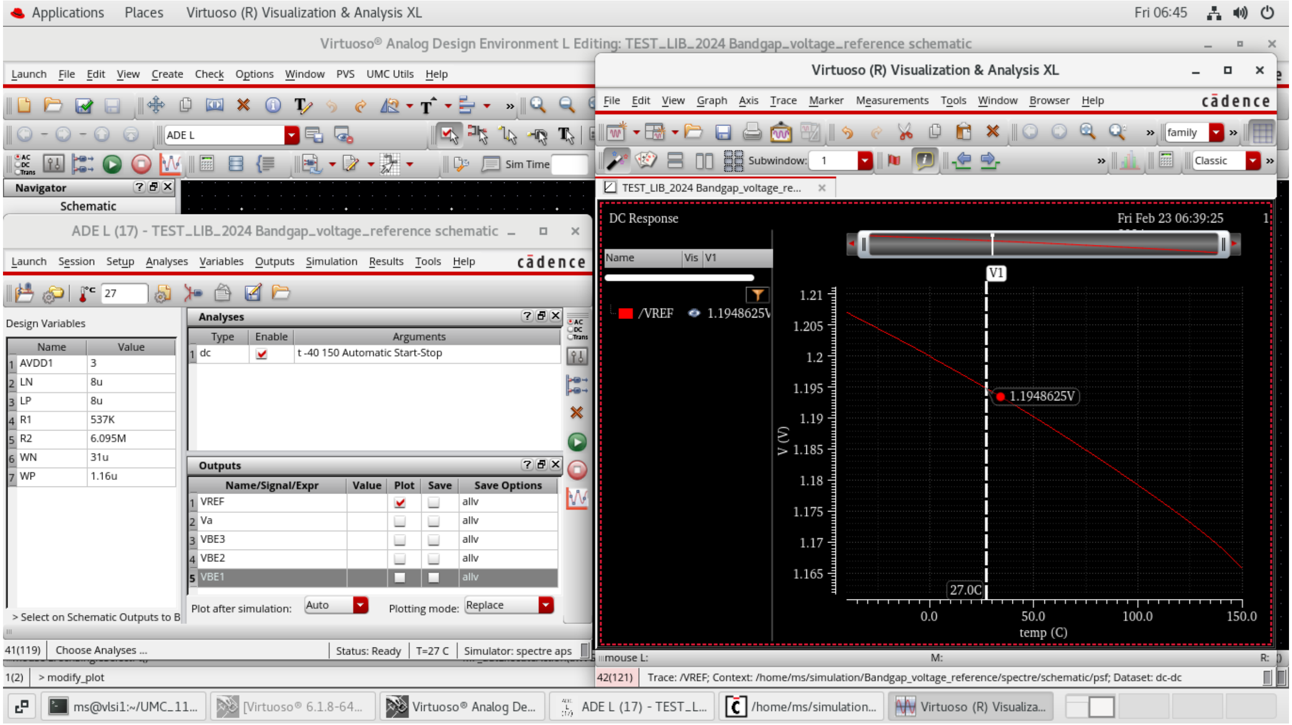
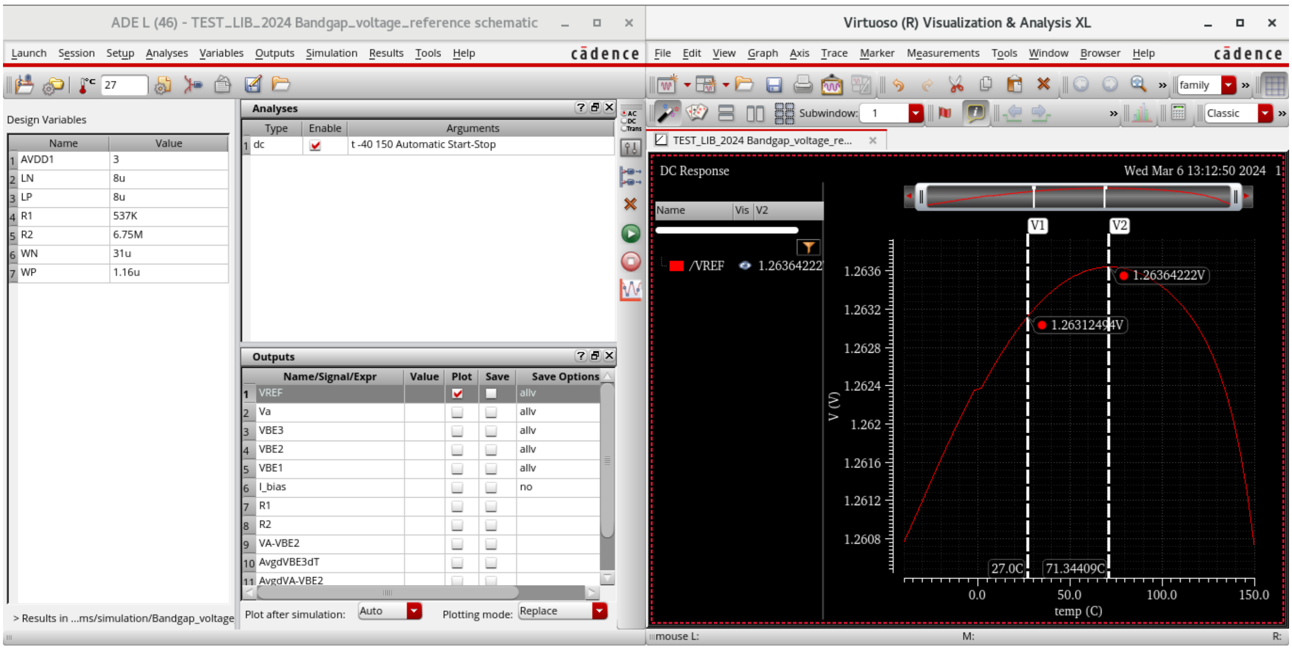


Given: AVDD = 3V, Ibias = 100 nA
Calculated: R1 = 0.537 MOhm, R2 = 6.095 Mohm, WP=1.16um, WN = 0.31um

For dVREF/dT = 0, R2 is changed to **6.75 Mohm**.



Corners – C0 (SS), C1 (TT), C2 (FF)

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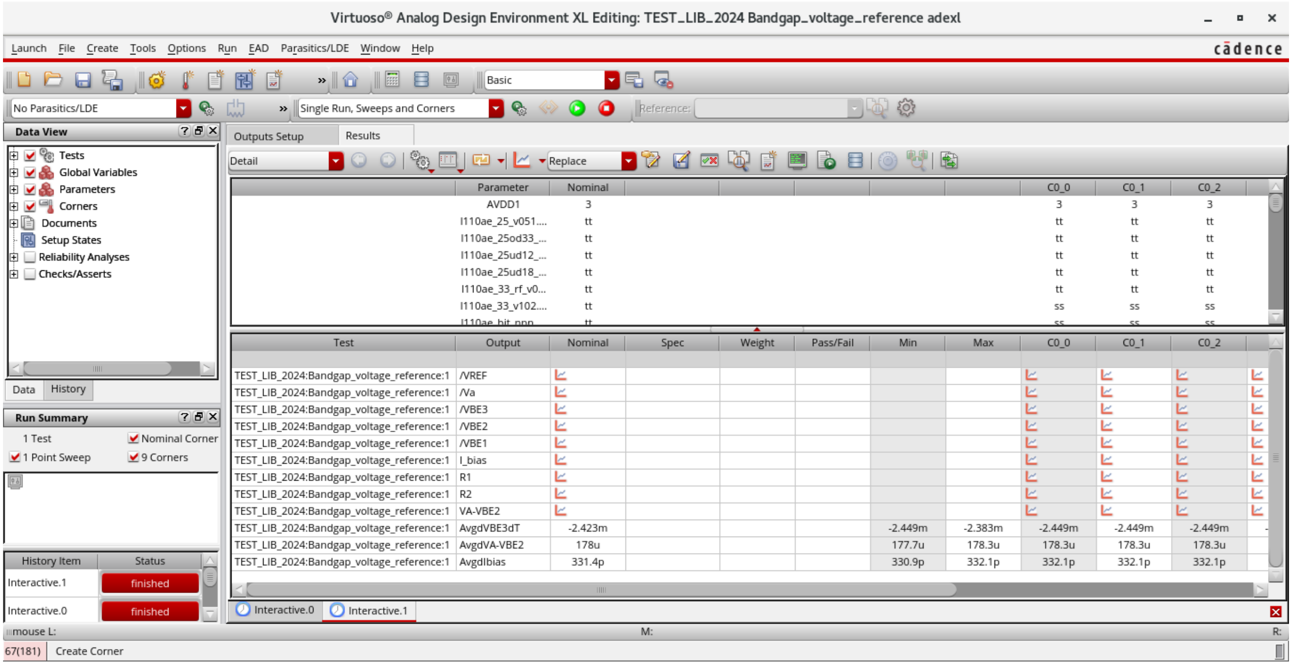

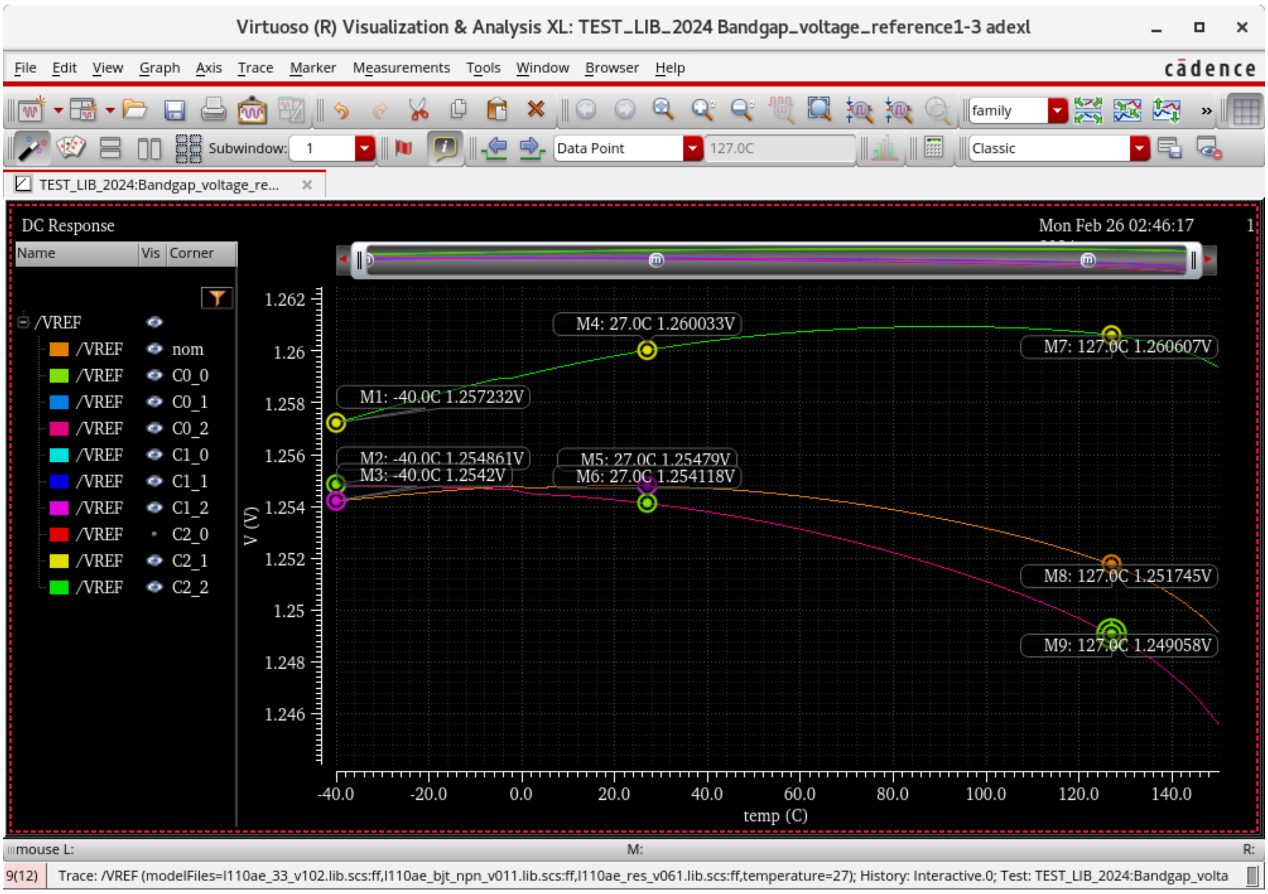

<u>VREF vs Temperature</u>

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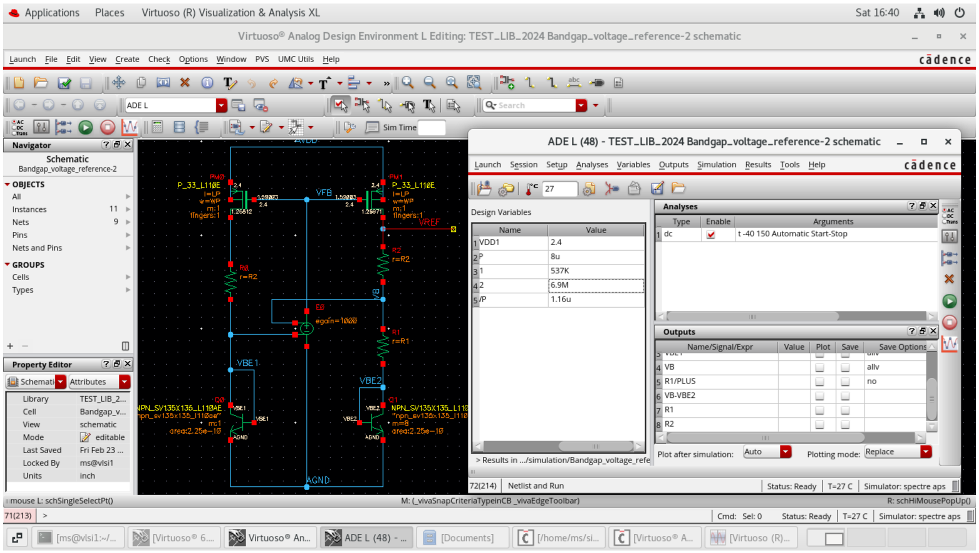
Corners		SS			TT			FF		
AVDD		2.4V	3.0V	3.6V	2.4V	3.0V	3.6V	2.4V	3.0V	3.6V
VBE3 (V)	@27C	0.504	0.5045	0.505	0.4968	0.4973	0.497	0.4937	0.4943	0.4948
avg(dVBE3/dT)	V/C	0.002448	0.002449	0.002449	0.002423	00.002423	0.002423	0.002383	0.002383	0.002383
VA-VBE2	V	0.05503	0.05597	0.05691	0.05517	0.05612	0.05707	0.05533	0.0563	0.05726
avg(d(VA - VBE2)/dT)	V/C	0.0001786	0.0001783	0.000178	0.0001791	0.000178	0.0001782	0.0001783	0.0001777	0.0001779
R1	Ohm	5.45M	537000	5.3M	5372000	537000	5219000	5329000	537000	5176000
R2	Ohm	1772000	6712000	1971000	1852000	6711000	2015000	1893000	6710000	2095000
Ibias1(A)	@27C	0.102u	0.104u	0.106u	0.102u	0.104u	0.106u	0.103u	0.104u	0.106u
Avg (dIbias1/dT)	A/C	0.333n	0.332n	0.331n	0.332n	0.331n	0.332n	0.332n	0.331n	0.331n
VREF1	@27C	1.244	1.26	1.276	1.239	1.255	1.27	1.238	1.254	1.27
VREF1_Max	V	1.244	1.259	1.277	1.237	1.254	1.271	1.237	1.255	1.272
VREF1_Min	V	1.24	1.257	1.274	1.233	1.249	1.266	1.23	1.246	1.262
VREF1_Max - VREF1_Min	V	0.004	0.002	0.003	0.004	0.005	0.005	0.007	0.009	0.01
VREF1 temp co	ppm/C	16.92	8.36	12.36	17.019	20.98	20.70	29.78	37.74	41.37

Question – 3:-

Given:- AVDD = 3V, Ibias = 100n A, Vov = 0.2 V,
Calculated:- R3=R1 = 0.537 MOhm, R4 = R2 = 6.095 MOhm, LP = 8um, WP = 1.16um.



For dVREF/dT = 0, R2 is changed to **7.02 Mohm.**

Corners – C0 (SS), C1 (TT), C2 (FF)

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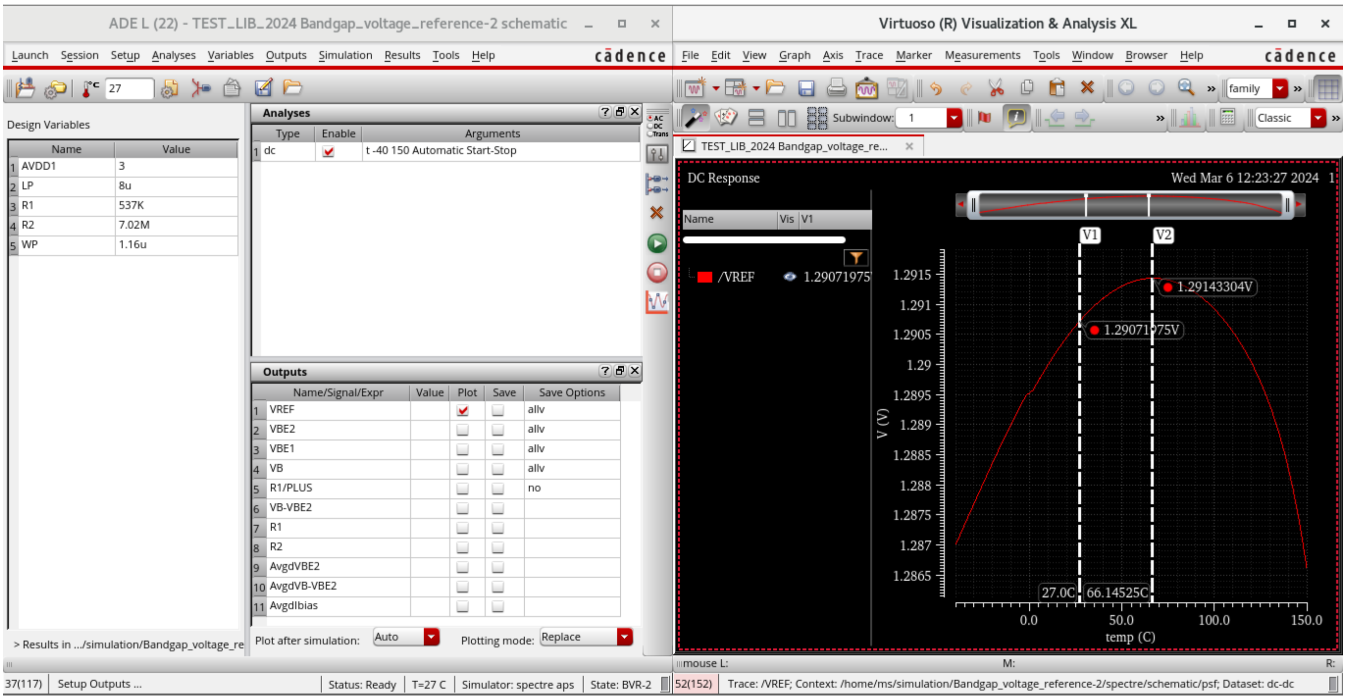

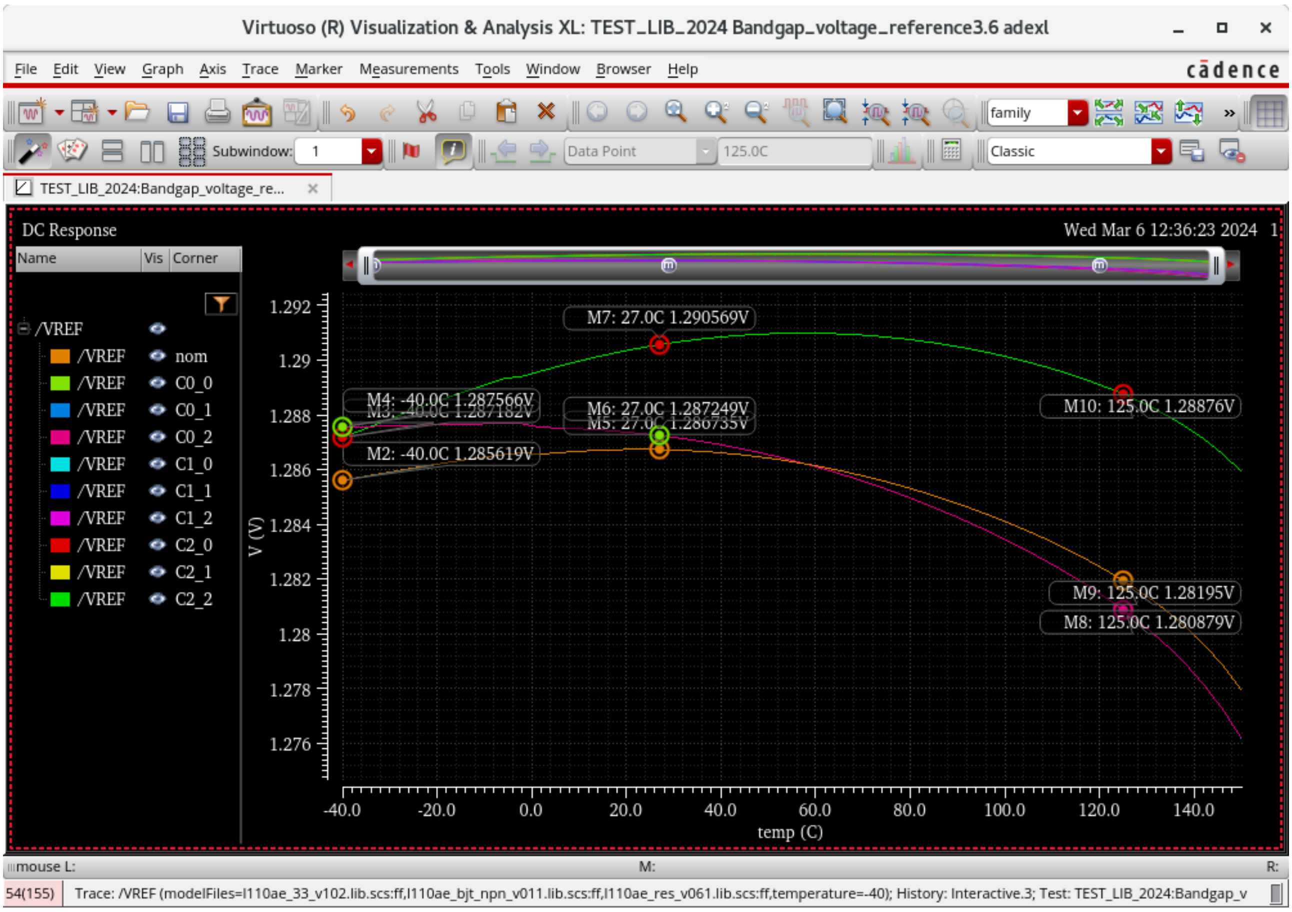

<u>VREF vs Temperature</u>

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Corners		SS			TT			FF		
AVDD		2.4V	3.0V	3.6V	2.4V	3.0V	3.6V	2.4V	3.0V	3.6V
VBE5	@27C	0.5042	0.5044	0.5047	0.497	0.4973	0.497	0.494	0.4943	0.4946
avg(dVBE5/dT)	V/C	0.002426	0.002452	0.002452	0.002452	0.002426	0.002426	0.00238	0.002386	0.002386
VB-VBE5	V	0.05555	0.05615	0.05675	0.05579	0.0563	0.0569	0.05604	0.05639	0.05723
avg(d(VB-VBE5)/dT	V/C	172.8u	172.8u	172.8u	172.3u	172.3u	172.3u	171.9u	171.8u	171.9u
R1	Ohm	537000	537000	537000	537000	537000	537000	537000	537000	537000
R2	Ohm	6.9M	6.9M	6.9M	6.9M	6.9M	6.9M	6.9M	6.9M	6.9M
Ibias	@27C	0.103u	0.104u	0.105u	0.103u	0.105u	0.106u	0.104u	0.105u	0.106u
Avg (dIbias/dT)	A/C	321.9p	321p	321.8p	320.9p	321p	320.9p	320p	320p	320p
VREF(V)	@27C	1.273	1.282	1.291	1.27	1.278	1.287	1.27	1.279	1.287
VREF_Max (V)		1.27	1.279	1.287	1.268	1.277	1.286	1.27	1.279	1.288
VREF_Min (V)		1.269	1.277	1.286	1.261	1.269	1.278	1.259	1.268	1.276
(VREF_Max-VREF_Min) (V)		0.001	0.002	0.001	0.007	0.008	0.008	0.011	0.011	0.012
VREF Temp co	ppm/C	4.144	8.23	4.089	29.05	32.97	32.74	45.58	45.26	49.035